2017 年真题参考答案

一、选择题

(1) A. (2) D. (3) C. (4) C. (5) A. (6) B. (7) C. (8) B.

二、填空题

(9) $\frac{\pi^3}{2}$. (10) $\left(C+\frac{1}{2}t\right)2^t$. (11) $1+\mathrm{e}^{-Q}-Q\mathrm{e}^{-Q}$. (12) $xy\mathrm{e}^{y}$. (13) 2. (14) $\frac{9}{2}$.

三、解答题

(15) $\frac{2}{3}$.

(16) $\frac{(2-\sqrt{2})\pi}{16}$.

(17) $\frac{1}{4}$.

(18) $\left(\frac{1}{\ln 2}-1,\frac{1}{2}\right)$.

(19) (Ⅰ) 证明略.

(Ⅱ) 证明略. $S(x)=\frac{\mathrm{e}^{-x}}{1-x}$, $x\in(-1,1)$.

(20) (Ⅰ) 证明略.

(Ⅱ) $k(1,2,-1)^{\mathrm{T}}+(1,1,1)^{\mathrm{T}}$ 为线性方程组 $\boldsymbol{Ax}=\boldsymbol{\beta}$ 的通解,其中 k 为任意常数.

(21) $a=2$, $\boldsymbol{Q}=\begin{pmatrix}-\frac{1}{\sqrt{2}} & \frac{1}{\sqrt{3}} & \frac{1}{\sqrt{6}}\\ 0 & -\frac{1}{\sqrt{3}} & \frac{2}{\sqrt{6}}\\ \frac{1}{\sqrt{2}} & \frac{1}{\sqrt{3}} & \frac{1}{\sqrt{6}}\end{pmatrix}$,且 $\boldsymbol{Q}^{\mathrm{T}}\boldsymbol{AQ}=\begin{pmatrix}6 & 0 & 0\\ 0 & -3 & 0\\ 0 & 0 & 0\end{pmatrix}$.

(22) (Ⅰ) $\frac{4}{9}$.

(Ⅱ) $f_Z(z)=\begin{cases}z, & 0<z<1,\\ z-2, & 2<z<3,\\ 0, & \text{其他.}\end{cases}$

(23) (Ⅰ) $f_{Z_1}(z)=\begin{cases}\sqrt{\frac{2}{\pi}}\frac{1}{\sigma}\mathrm{e}^{-\frac{z^2}{2\sigma^2}}, & z\geqslant 0,\\ 0, & z<0.\end{cases}$

(Ⅱ) $\hat{\sigma}=\sqrt{\frac{\pi}{2}}\,\overline{Z}$.

(Ⅲ) $\hat{\sigma}=\sqrt{\frac{1}{n}\sum_{i=1}^{n}Z_i^2}$.